Assignment 3: Longitudinal Text Messaging Analysis

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Introduction

This report utilizes the TextMessages.csv dataset to examine the relationship between two groups of participants who had their texting habits recorded at two different points in time. The variables `Baseline` and `Six_Months` represent the total number of text messages a person typed initially as well as six months later.

Findings show that Group 2 had a higher average number of baseline texts sent than Group 1, of 65.6 versus 64.84. However, Group 1 had the greatest change in frequency and ultimately a lower average texting rate after six months than Group 2 (52.96 versus 61.84).

Section 1: Environment Setup & Data Preparation

First, I loaded the `pastecs`, `tidyr`, and `dplyr` packages to perform data wrangling and descriptive statistics. Then, I imported `TextMessages.csv` from the project repository.

I converted `Group` into a labeled factor and the dataset into a tidy long format using `pivot_longer()` so that `Time` is a categorical variable with `Group`. Then, I exported this clean long dataset as `TextMessages_Clean_Long.csv` so my partner can import it directly to create the required faceted boxplots and bar charts.

```
# I loaded pastecs to compute detailed descriptive statistics using stat.desc()
library(pastecs)

# I loaded tidyr and dplyr to perform data wrangling and reshape the dataset
library(tidyr)
library(dplyr)

# I imported the shared dataset from the project repository into text_data
text_data <- read.csv("TextMessages.csv")

# I checked the initial wide structure and dimension of the raw data
str(text_data)
```

```
## 'data.frame':    50 obs. of  4 variables:
##  $ Group      : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ Baseline   : int  52 68 85 47 73 57 63 50 66 60 ...
##  $ Six_months : int  32 48 62 16 63 53 59 58 59 57 ...
##  $ Participant: int  1 2 3 4 5 6 7 8 9 10 ...
```

```
head(text_data)
```

```
##   Group Baseline Six_months Participant
## 1     1       52         32           1
## 2     1       68         48           2
```

```
## 3     1       85         62           3
## 4     1       47         16           4
## 5     1       73         63           5
## 6     1       57         53           6
```

```
# I converted Group into a labeled factor for clean categorical identification
text_data$Group <- factor(text_data$Group,
                          levels = c(1, 2),
                          labels = c("Group 1", "Group 2"))

# I reshaped Baseline and Six_months from wide into tidy long format
text_long <- text_data %>%
  pivot_longer(
    cols = c(Baseline, Six_months),
    names_to = "Time",
    values_to = "Text_Messages"
  ) %>%
  mutate(
    Time = factor(Time,
                  levels = c("Baseline", "Six_months"),
                  labels = c("Baseline", "6 Months"))
  )

# I verified the structure of the reshaped long data frame
str(text_long)
```

```
## tibble [100 x 4] (S3: tbl_df/tbl/data.frame)
##  $ Group        : Factor w/ 2 levels "Group 1","Group 2": 1 1 1 1 1 1 1 1 1 1 ...
##  $ Participant  : int [1:100] 1 1 2 2 3 3 4 4 5 5 ...
##  $ Time         : Factor w/ 2 levels "Baseline","6 Months": 1 2 1 2 1 2 1 2 1 2 ...
##  $ Text_Messages: int [1:100] 52 32 68 48 85 62 47 16 73 63 ...
```

```
head(text_long)
```

```
## # A tibble: 6 x 4
##   Group   Participant Time     Text_Messages
##   <fct>         <int> <fct>            <int>
## 1 Group 1           1 Baseline            52
## 2 Group 1           1 6 Months            32
## 3 Group 1           2 Baseline            68
## 4 Group 1           2 6 Months            48
## 5 Group 1           3 Baseline            85
## 6 Group 1           3 6 Months            62
```

```
# I saved the clean long data frame so my partner can use it for visual tasks
write.csv(text_long, "TextMessages_Clean_Long.csv", row.names = FALSE)
```

Section 2: Stratified Summary Statistics by Group and Time

I calculated summary statistics for `Text_Messages` stratified jointly by `Group` and `Time` using `by()` paired with `stat.desc()` from the `pastecs` package.

```
# I computed stratified summary statistics across Group and Time using stat.desc
stratified_summary <- by(
```

```
  text_long$Text_Messages,
  list(text_long$Group, text_long$Time),
  stat.desc,
  basic = TRUE
)

# I displayed the stratified summary statistics for Group 1 and Group 2 across both time points
stratified_summary
```

```
## : Group 1
## : Baseline
##      nbr.val     nbr.null       nbr.na          min          max        range
##    25.000000     0.000000     0.000000    47.000000    85.000000    38.000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
##  1621.000000    64.000000    64.840000     2.135946     4.408377   114.056667
##      std.dev     coef.var
##    10.679732     0.164709
## ------------------------------------------------------------
## : Group 2
## : Baseline
##      nbr.val     nbr.null       nbr.na          min          max        range
##   25.0000000    0.0000000    0.0000000   46.0000000   89.0000000   43.0000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
## 1640.0000000   65.0000000   65.6000000    2.1671794    4.4728385  117.4166667
##      std.dev     coef.var
##   10.8358971    0.1651814
## ------------------------------------------------------------
## : Group 1
## : 6 Months
##      nbr.val     nbr.null       nbr.na          min          max        range
##   25.0000000    0.0000000    0.0000000    9.0000000   78.0000000   69.0000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
## 1324.0000000   58.0000000   52.9600000    3.2662313    6.7411700  266.7066667
##      std.dev     coef.var
##   16.3311563    0.3083678
## ------------------------------------------------------------
## : Group 2
## : 6 Months
##      nbr.val     nbr.null       nbr.na          min          max        range
##   25.0000000    0.0000000    0.0000000   46.0000000   79.0000000   33.0000000
##          sum       median         mean      SE.mean CI.mean.0.95          var
## 1546.0000000   62.0000000   61.8400000    1.8820910    3.8844450   88.5566667
##      std.dev     coef.var
##    9.4104552    0.1521742
```

Summary:

Group 1 Statistics: * **Baseline (n = 25):** Mean = 64.84, Median = 64.00, SD = 10.68, Variance = 114.06, Range = 38.00 (Minimum = 47.00, Maximum = 85.00), SE = 2.14. * **6 Months (n = 25):** Mean = 52.96, Median = 58.00, SD = 16.33, Variance = 266.71, Range = 69.00 (Minimum = 9.00, Maximum = 78.00), SE = 3.27.

Group 2 Statistics: * **Baseline (n = 25):** Mean = 65.60, Median = 65.00, SD = 10.84, Variance = 117.42, Range = 43.00 (Minimum = 46.00, Maximum = 89.00), SE = 2.17. * **6 Months (n = 25):** Mean =

61.84, Median = 62.00, SD = 9.41, Variance = 88.56, Range = 33.00 (Minimum = 46.00, Maximum = 79.00), SE = 1.88.

Inferences:

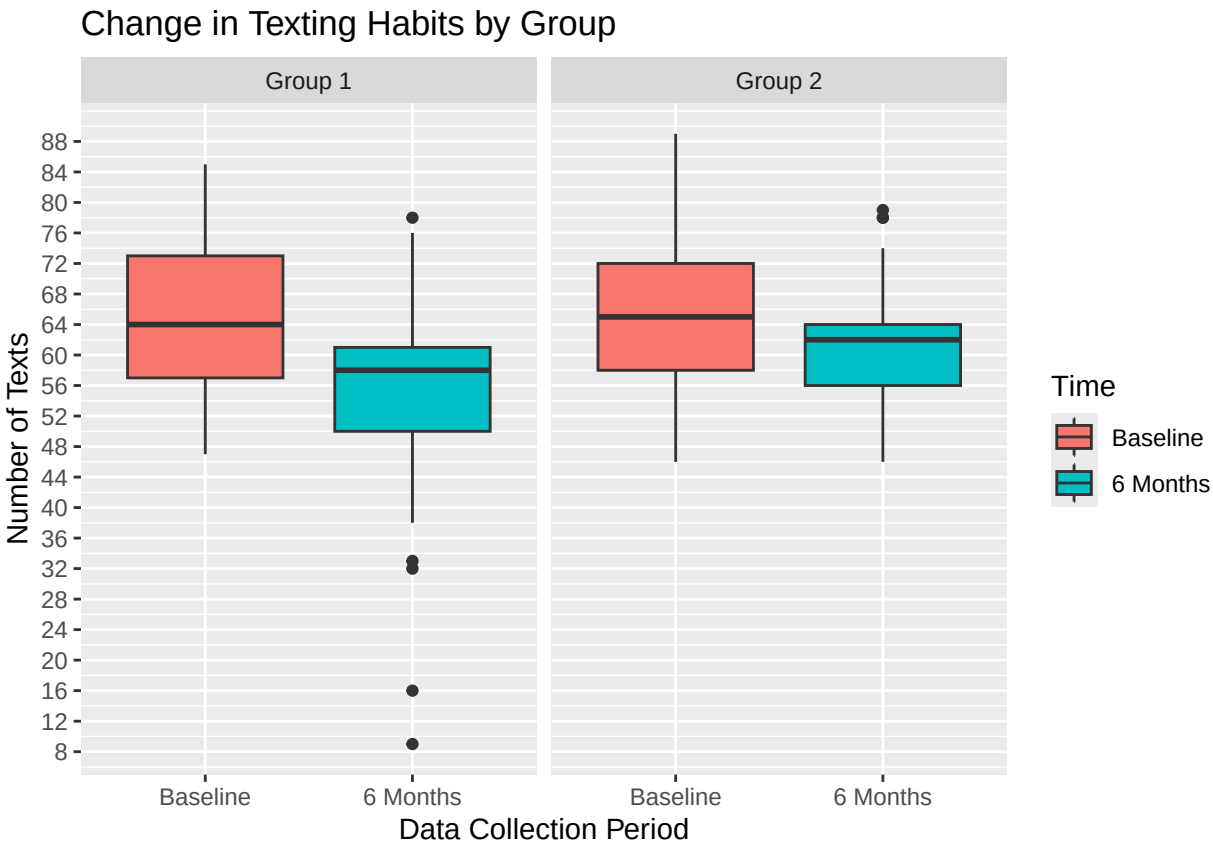
1. **Baseline:** At Baseline, both intervention groups exhibited nearly identical messaging rates (Group 1 Mean = 64.84 vs. Group 2 Mean = 65.60) and very similar variability (SD = 10.68 vs. 10.84), showing that the groups were comparable prior to intervention.
2. **Longitudinal Change:** Over the 6-month period, Group 1 demonstrated a substantial decrease in text messaging, dropping by an average of 11.88 messages (from 64.84 down to 52.96). In contrast, Group 2 showed a much smaller reduction of only 3.76 messages (from 65.60 down to 61.84).
3. **Variance:** While Group 1 experienced a greater reduction in text messages, its spread widened considerably at 6 months (SD increased from 10.68 to 16.33; range increased from 38 to 69). In contrast, Group 2 maintained a tight, consistent spread (SD decreased from 10.84 to 9.41; range decreased from 43 to 33).

Section 3: Boxplot Visualization by Group and Time

```
#The ggplot2 package is first loaded to construct visualizations
library(ggplot2)

#To construct a boxplot that not only shows the change in texting habits over a
#period of six months, but also shows the difference between Group 1 and Group
#2's change in texting, a faceted approach is used.
boxplot_text <- text_long %>% ggplot(aes(Time, Text_Messages, fill=Time))+
  geom_boxplot()+
  labs(title="Change in Texting Habits by Group", x="Data Collection Period",
       y="Number of Texts")+
  scale_y_continuous(breaks=seq(from=0, to=90, by=4))+
  facet_wrap(Group~.)

#To view the faceted boxplot, the object is re-typed.
boxplot_text
```



From this, we are able to see that Group 2 had both the highest and lowest baseline texting scores, as well as the highest score after six months, while Group 1 had the lowest score after six months. Group 2 has a baseline score that appears to be approximately normally distributed, whereas Group 1 appears to have a slightly right skewed distribution. However, both follow-up scores are noticeably left skewed. Both groups had a noticeable decrease in the number of text messages sent after six months, with Group 1 appearing to have a greater decrease in median scores (from 64 to 58 versus 65 to 62). Althought Group 1 has a slightly lower median baseline score than Group 2, its interquartile range is slightly greater, suggesting a greater spread throughout the middle 50% of the data. Further, Group 1 had considerably low outliers in its six month scores, whereas Group 2 had multiple high outliers. Finally, both groups have a much smaller interquartile range at six months compared to their baseline scores, consistent with a smaller amount of variability among the middle 50% of follow-up scores.

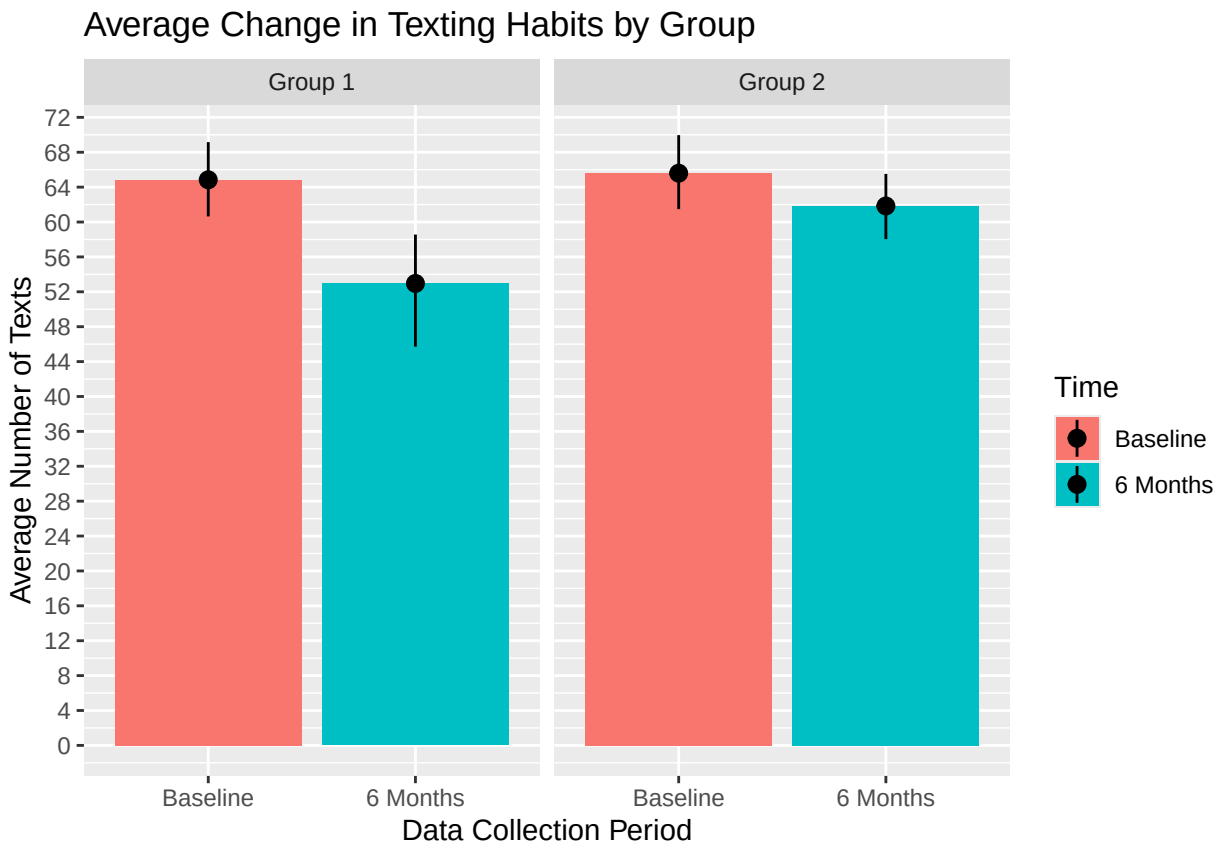
Section 4: Bar Chart Visualization by Group and Time

```
#The Hmisc package is loaded to construct errorbars that represent the 95%
#confidence interval of the mean.
library(Hmisc)

#Similarly, to construct a bar graph that displays a side-by-side comparison of
#the average baseline score and the average six month score for Groups 1 and 2,
#a faceted approach is used once more. An error bar is also included to depict
#the 95% confidence interval for each score's average.
barchart_text <- text_long %>% ggplot(aes(Time, Text_Messages, fill=Time))+
  stat_summary(fun=mean, geom="bar")+
  facet_wrap(Group~.)+
```

```
  scale_y_continuous(breaks=seq(from=0, to=100, by=4))+
  stat_summary(fun.data = mean_cl_boot, geom= "pointrange", colour="Black")+
  labs(title="Average Change in Texting Habits by Group",
       x="Data Collection Period", y="Average Number of Texts")

#To view the bar chart, the object is re-typed.
barchart_text
```



Beginning with the baseline scores, we are able to see that Group 1 has a slightly lower average score than Group 2, of approximately 64.84 versus 65.60, consistent with the previously examined summary statistics. Group 2 appears to have a slightly larger 95% confidence interval than Group 1. Given that both groups have the same sample size, this can be attributed to Group 1 having a lower standard deviation than Group 2, of 10.68 versus 10.84, respectively. Both groups exhibit a decrease in average texting scores after six months, with Group 1 not only having a smaller average follow-up score than Group 2 (of 52.96 versus 61.84), but also a much greater decrease in average scores of approximately 12 versus 4. Unlike the baseline scores, we can see that the six month scores have a larger difference between the two 95% confidence intervals, with Group 1 having a greater distribution of scores about the mean than Group 2. From this, we can conclude that Group 1 also had a greater change in variability than Group 2 after six months.

Conclusion

Overall, Group 1 had a greater average change of texting habits within six months than Group 2. Although Group 1 had a lower average baseline score than Group 2, the difference of 64.84 versus 65.60 was marginal. Further, comparing each group's mean to its median showed that both groups appeared to be roughly symmetric, with Group 1 having a slight right skew and Group 2 appearing approximately normally distributed.

After six months, however, the two groups did not appear to have similar averages or distributions. Group 1 now had an average texting rate of 52.96 messages, which was much lower than Group 2's average of 61.84. Initially, Group 1 had a smaller range and standard deviation than Group 2, of 38 & 10.68 versus 43 & 10.84. After six months, both the visualizations and summary statistics show that Group 1 now has a considerably larger range and standard deviation than Group 2, of 69 and 16.33 versus 33 and 9.41. Lastly, comparing the means and medians of the Six Month measurements shows that both groups are now left skewed.

These findings illustrate that although the two groups had relatively similar texting habits initially, they diverged the six month period, with Group 1 experiencing a more drastic decrease and greater dispersion, while Group 2 maintained a more consistent decline in texting frequency.